

Smart Disaster Response MIS

MIS Reports & Dashboards

DELIVERABLE — REPORTING & ANALYTICS

Course: Database Systems

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1. Executive Analytics Dashboard

The **Executive Dashboard** provides high-level situational awareness for senior decision-makers, integrating data from all system modules into a unified interface for rapid assessment of ongoing disaster conditions, resource health, and organisational capacity.

Visualisation	Chart Type	Purpose
Incident Heatmap	Bar Chart (Recharts)	Active disaster breakdown by type: Flood, Earthquake, Fire
Severity Distribution	Pie Chart	Ratio of Critical vs. Low severity — for resource prioritisation
Trend Analysis	Line Chart	Daily report volume over 30 days to identify peaking events
Financial Health	Balance Sheet Panel	Real-time total donations vs. expenses and current liquidity

Table 1 — Executive Dashboard visualisations

2. Operational Dashboards

Role-specific dashboards ensure users see only data relevant to their function, reducing cognitive load during high-pressure scenarios.

2.1 Emergency Operator Dashboard

- **Real-Time Queue:** All 'Pending' reports with an elapsed-time counter, enabling prioritisation of long-waiting incidents.
- **Team Status Map:** Available vs. assigned rescue teams alongside current member counts.
- **Quick Dispatch:** Single-click interface to link a pending report to an available team.

2.2 Logistics & Warehouse Dashboard

- **Stock Alert Panel:** Items where quantity_available is below threshold_level, sourced from vw_WarehouseInventorySummary.
- **Allocation Pipeline:** Resource requests tracked from Pending Approval to Dispatched to Consumed.

2.3 Medical & Hospital Dashboard

- **Bed Occupancy Report:** All active hospitals with real-time percentage occupancy, sourced from vw_HospitalCapacity.
- **Patient Triage:** Condition distribution — Stable, Serious, Critical — across all active facilities.

3. Automated Performance Reports

Automated reports support operational transparency, regulatory compliance, and post-incident analysis.

Report Name	Frequency	Target Audience	Data Source
Response Time Audit	Real-Time	System Admins	EmergencyReports (Resolution Time)
Inventory Turnover	Weekly	Warehouse Manager	ResourceAllocations
Financial Transparency	Monthly	Finance / Public	FinanceTransactions
User Activity Audit	On-Demand	Security / Admins	AuditLog

Table 2 — Automated report schedule and data sources

4. Conclusion

The MIS Dashboards ensure that data is not merely stored but rendered **actionable**. Role-specific views, executive analytics, and automated periodic reports empower the Disaster Response Management team to make **data-driven decisions** under the time pressure of active emergencies.

- Role-specific dashboards surface only what is operationally relevant to each user.
- Real-time views backed by indexed queries ensure sub-second refresh rates during peak incident volumes.
- Automated reports eliminate manual compilation and provide auditable records for post-incident review.